



MSpangenome: Simulating population pangenomes under coalescent demographic models

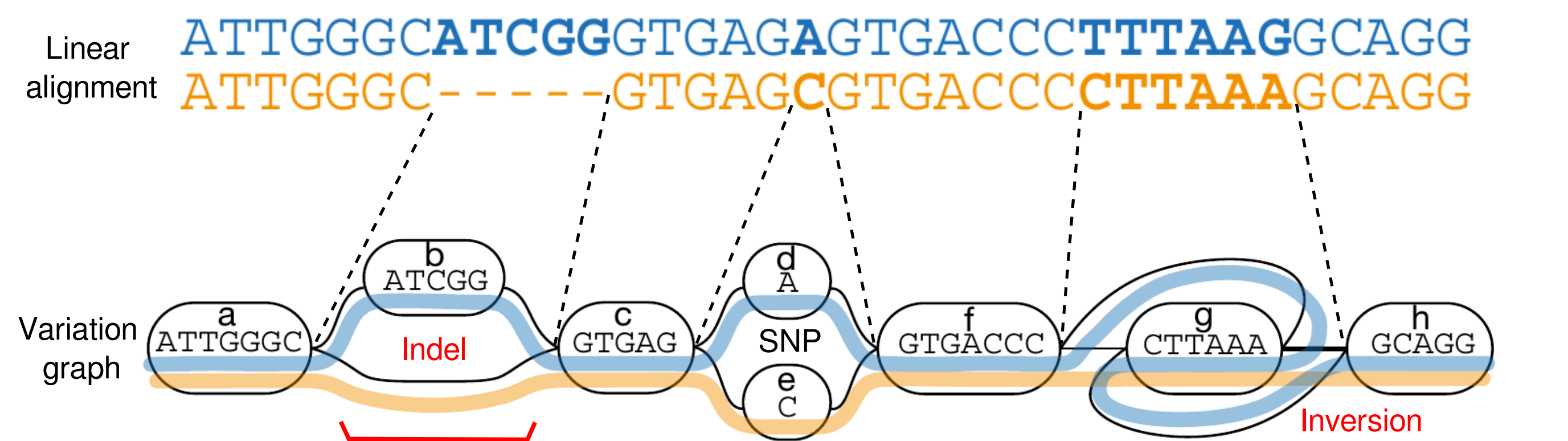
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Structural variants and pangenomes

Context: **Pangenome** variation graphs gain more and more interest to comprehensively represent genetic diversity including **structural variants**.



Structural variants (SVs): >50bp variations that change chromosome structure.

Adapted from Liao et al. (2022)

Problem : we don't know how to simulate realistic pangenomes

We present MSpangenome, a new tool for this purpose :

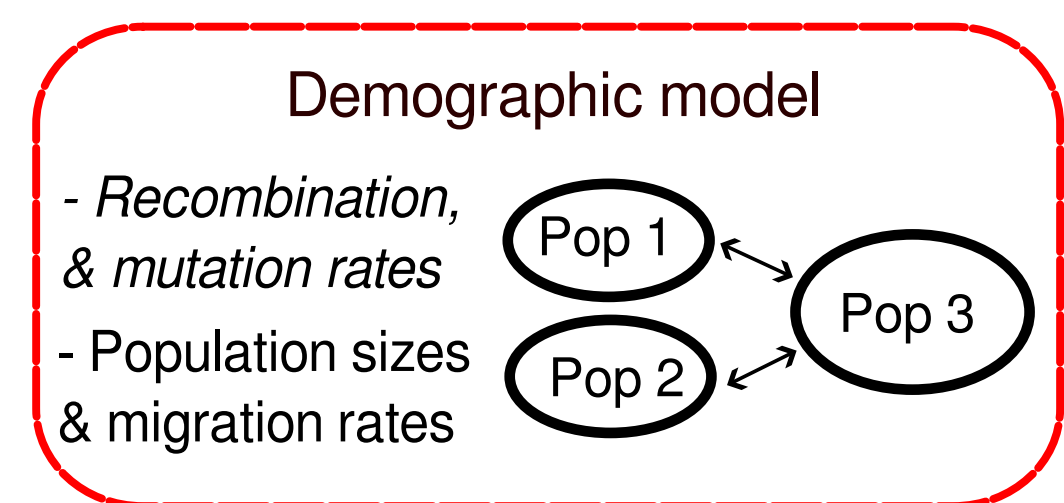
Features :

- Consider complex **demographic scenarios**
- Realistic (nested) **variants**
- Produce large genomes
- Export graph to GFA

Use cases :

- Tool benchmarking
- Genome evolutionary inferences
- OOP library to insert SVs into graph

Algorithm

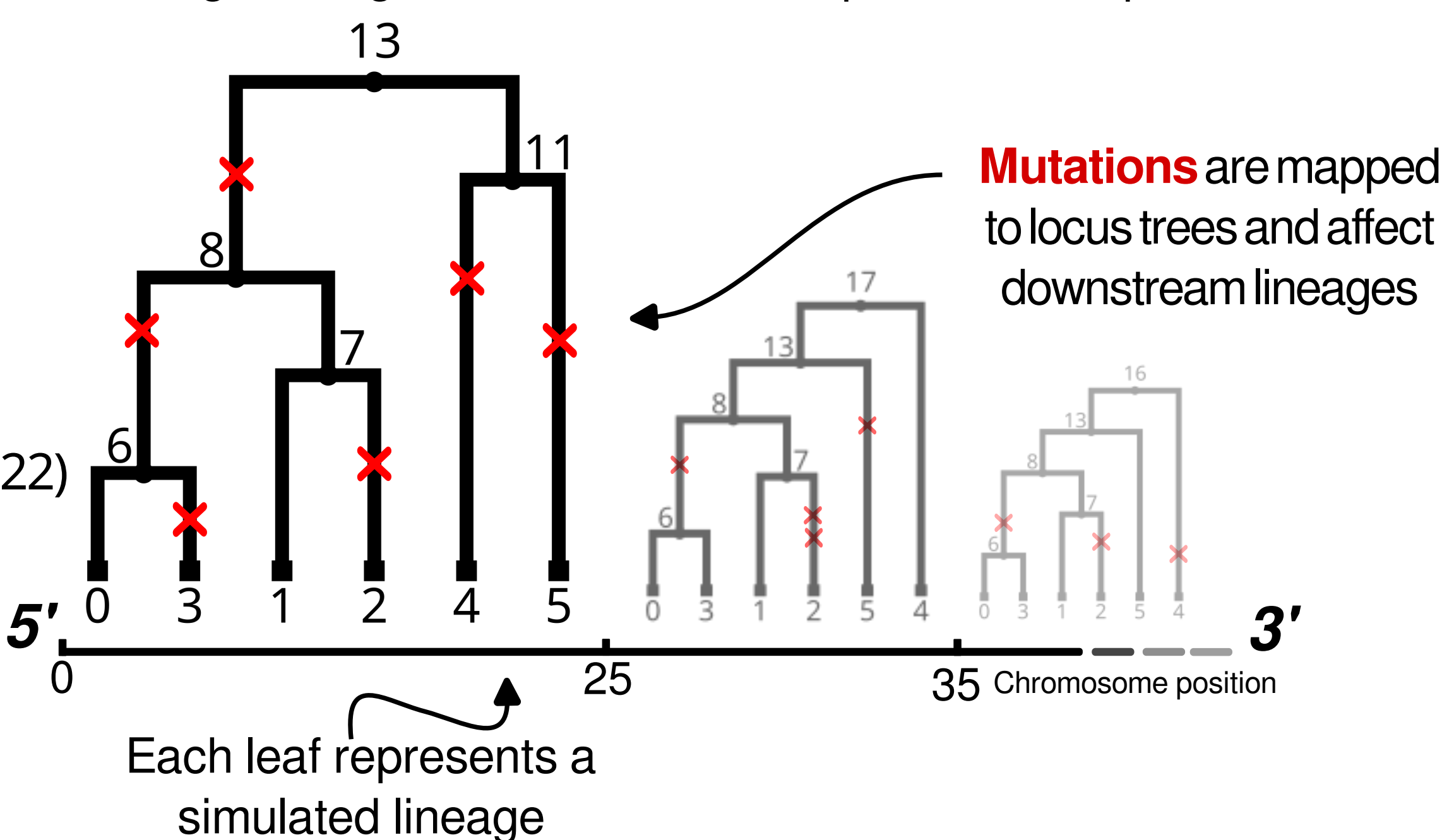


ms prime
MSprime Baumdicker et al. (2022)

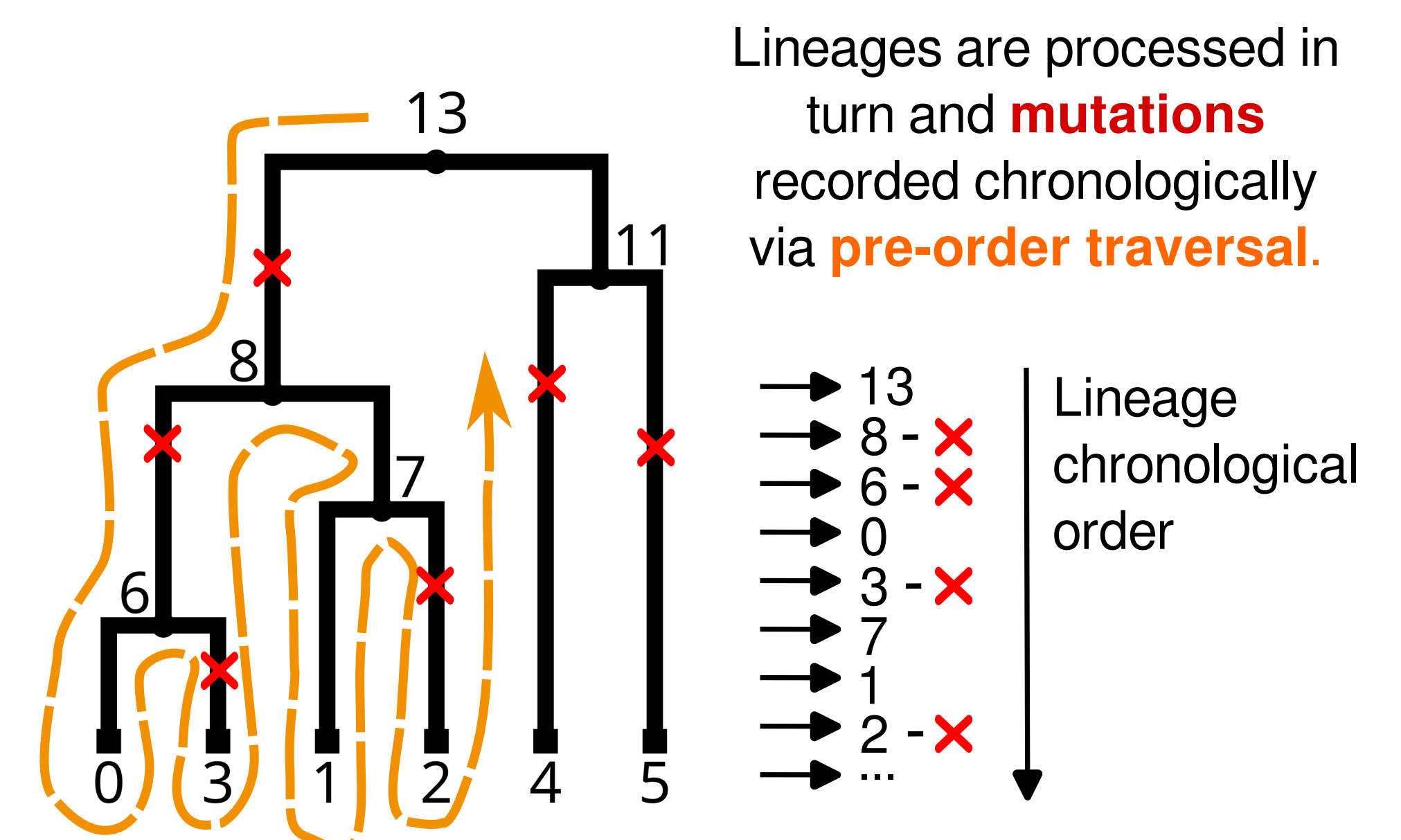
Produces genome genealogies

A. Ancestry simulation

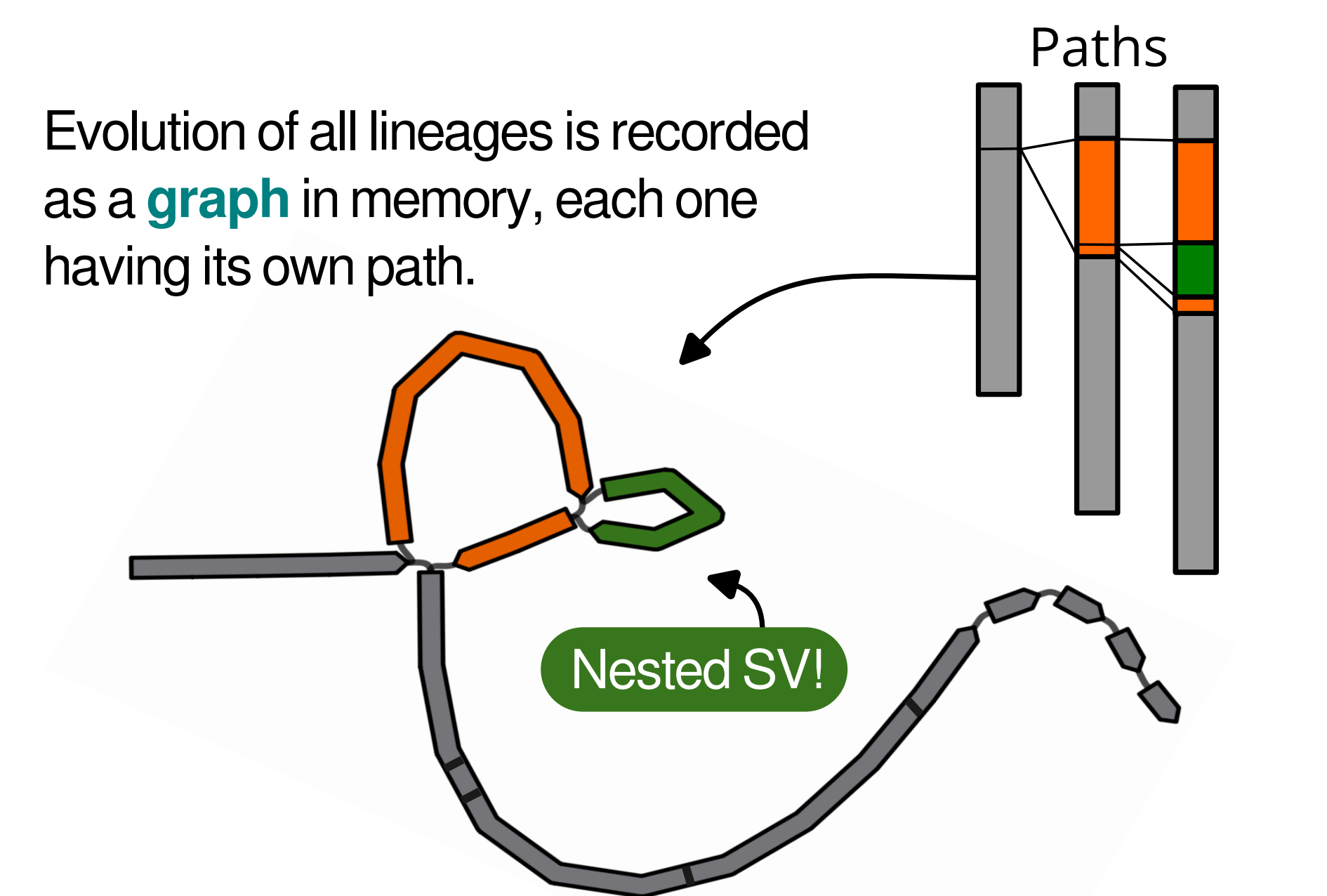
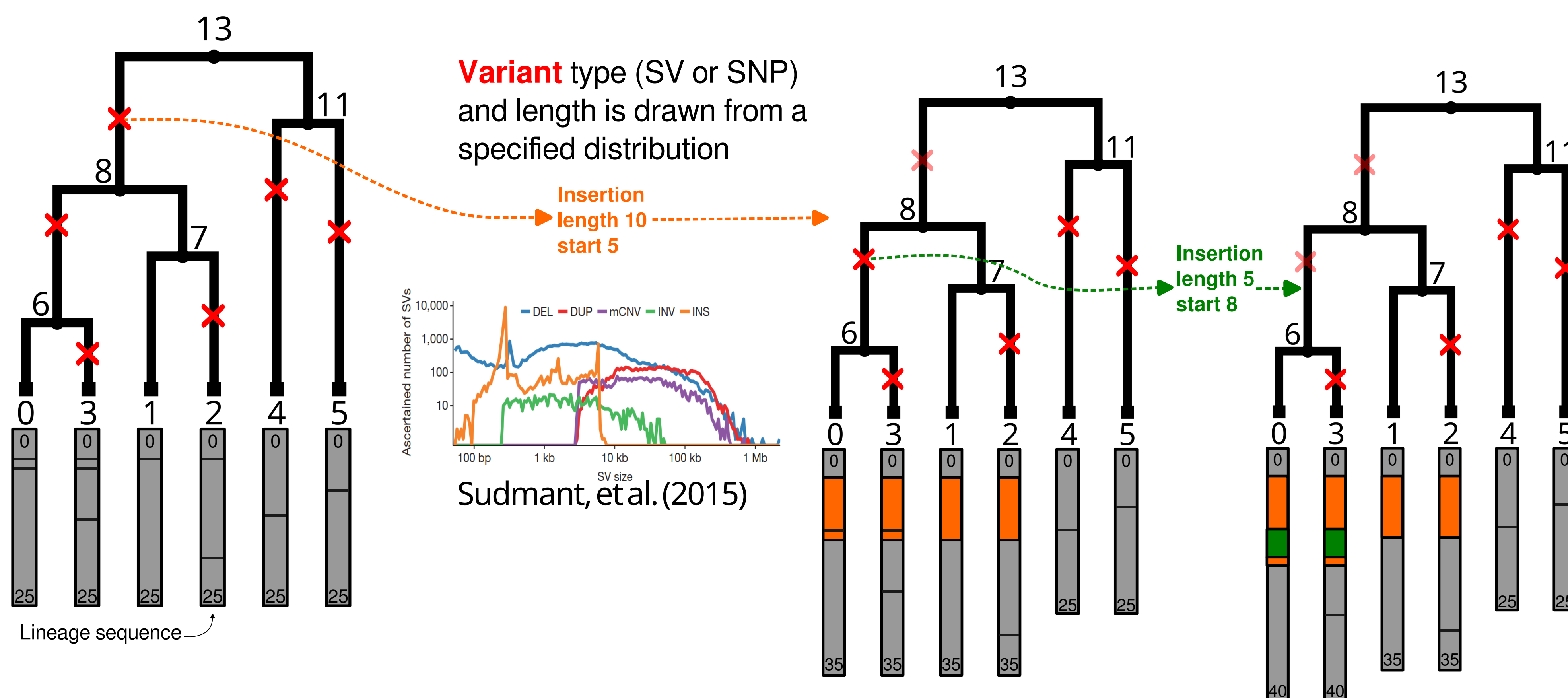
Along chromosomes, recombination blocks hold local genealogical trees that can be processed in parallel.



B. Recording mutations chronologically



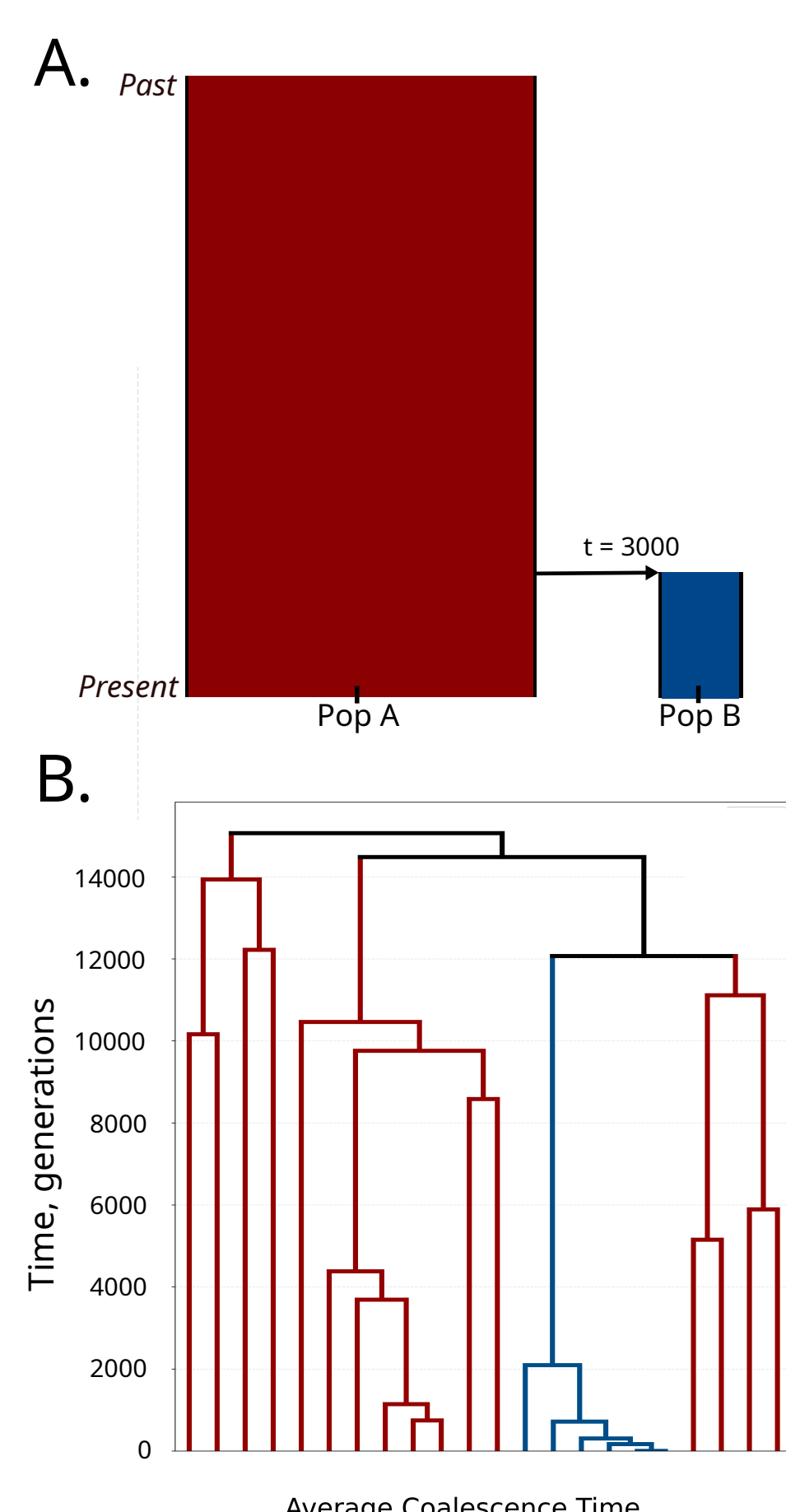
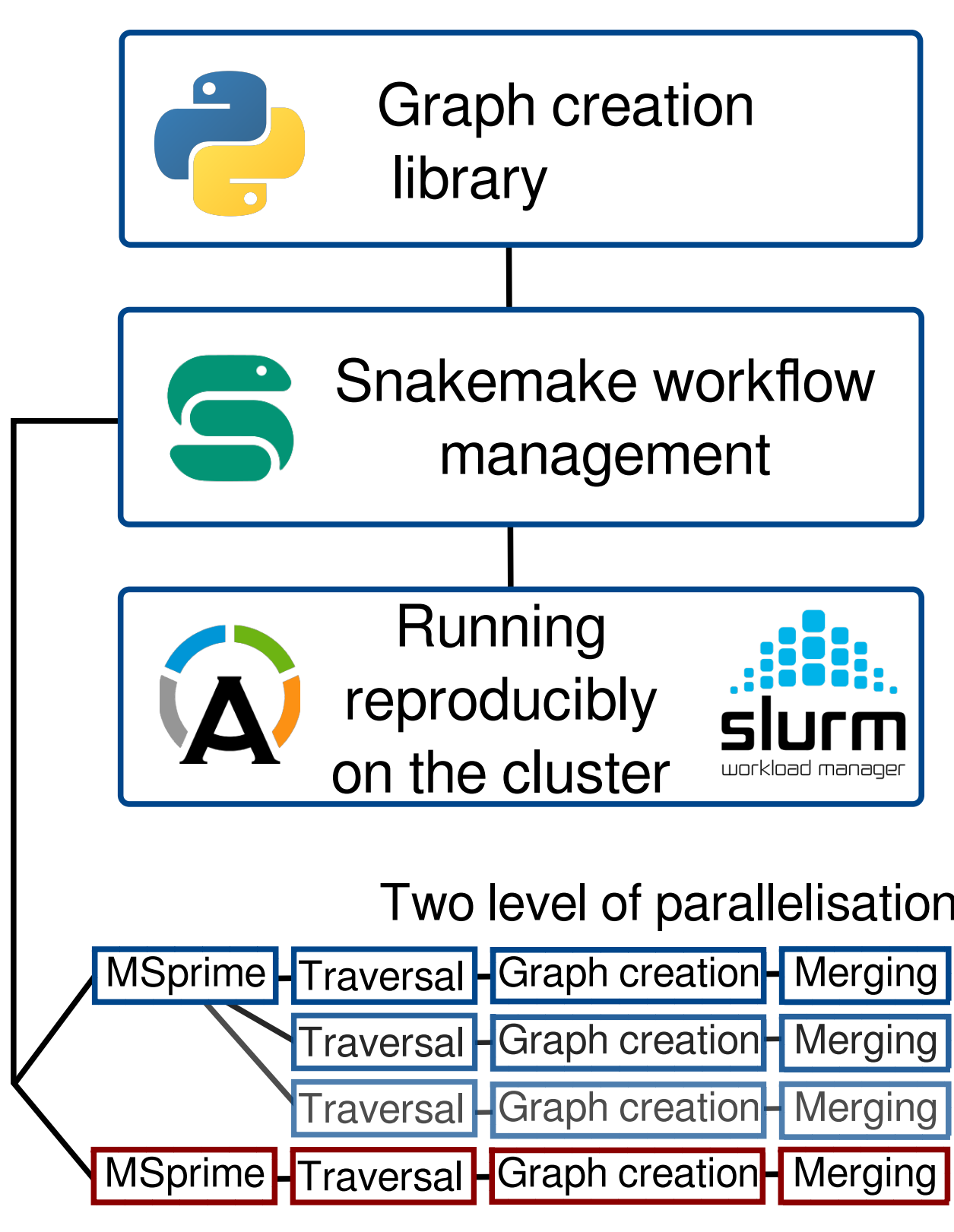
C. Adding structural variants with the MSpangenome library



D. Node merging and locus concatenation



MSpangenome overview



Simulation example

MSpangenome simulation of a *Staphylococcus aureus* pangenome, showing two colonies diverging 3000 generations ago.

